

1 Introduction

Difference-in-Differences (DiD) has become one of the most popular research designs used to evaluate causal effects of policy interventions. In its canonical format, there are two time periods and two groups: in the first period no one is treated, and in the second period some units are treated (the treated group), and some units are not (the comparison group). If, in the absence of treatment, the average outcomes for treated and comparison groups would have followed parallel paths over time (which is the so-called parallel trends assumption), one can estimate the average treatment effect for the treated subpopulation (ATT) by comparing the average change in outcomes experienced by the treated group to the average change in outcomes experienced by the comparison group. Methodological extensions of DiD methods often focus on this standard two periods, two groups setup; see, e.g., [Heckman et al. \(1997, 1998\)](#), [Abadie \(2005\)](#), [Athey and Imbens \(2006\)](#), [Qin and Zhang \(2008\)](#), [Bonhomme and Sauder \(2011\)](#), [de Chaisemartin and D’Haultfœuille \(2017\)](#), [Botosaru and Gutierrez \(2018\)](#), [Callaway et al. \(2018\)](#), and [Sant’Anna and Zhao \(2020\)](#).¹

Many DiD empirical applications, however, deviate from the canonical DiD setup and have more than two time periods and variation in treatment timing. In this article, we provide a unified framework for average treatment effects in DiD setups with multiple time periods, variation in treatment timing, and when the parallel trends assumption holds potentially only after conditioning on observed covariates. We concentrate our attention on DiD with staggered adoption, i.e., to DiD setups such that once units are treated, they remain treated in the following periods.

The core of our proposal relies on separating the DiD analysis into three separate steps: (i) identification of policy-relevant disaggregated causal parameters; (ii) aggregation of these parameters to form summary measures of the causal effects; and (iii) estimation and inference about these different target parameters. Our approach allows for estimation and inference on interpretable causal parameters allowing for arbitrary treatment effect heterogeneity and dynamic effects, thereby completely avoiding the issues of interpreting results of standard two-way fixed effects (TWFE) regressions as causal effects in DiD setups as pointed out by [Borusyak and Jaravel \(2017\)](#), [de Chaisemartin and D’Haultfœuille \(2020\)](#), [Goodman-Bacon \(2019\)](#), [Sun and Abraham \(2020\)](#), and [Athey and Imbens \(2018\)](#). In addition, it adds transparency and objectivity to the analysis ([Rubin \(2007, 2008\)](#)), and allows researchers to exploit a variety of estimation methods to answer different questions of interest.

The identification step of the analysis provides a blueprint for the other steps. In this paper, we pay particular attention to the disaggregated causal parameter that we call the *group-time average treatment effect*, i.e., the average treatment effect for group g at time t , where a “group” is defined by the time period when units are first treated. In the canonical DiD setup with two periods and two groups, these parameters reduce to the ATT which is typically the parameter of interest in that setup. An attractive feature of the group-time average treatment effect parameters is that they do not directly restrict heterogeneity with respect to observed covariates, the period in which units are first treated, or the evolution of treatment effects over time. As a consequence, these easy-to-interpret causal parameters can be directly used for learning about treatment effect heterogeneity, and/or to construct many other more aggregated causal

¹See Section 6 of [Athey and Imbens \(2006\)](#) and Theorem S1 in [de Chaisemartin and D’Haultfœuille \(2017\)](#) for notable exceptions that cover multiple periods and multiple groups.

Assumption 3 (Limited Treatment Anticipation). *There is a known $\delta \geq 0$ such that*

$$\mathbb{E}[Y_t(g)|X, G_g = 1] = \mathbb{E}[Y_t(0)|X, G_g = 1] \text{ a.s. for all } g \in \mathcal{G}, t \in \{1, \dots, \mathcal{T}\} \text{ such that } t < g - \delta.$$

Assumption 3 restricts anticipation of the treatment for all “eventually treated” groups. When $\delta = 0$, it imposes a “no-anticipation” assumption, see, e.g., [Abbring and van den Berg \(2003\)](#) and [Sianesi \(2004\)](#). This is likely to be the case when the treatment path is not a priori known and/or when units are not the ones who “choose” treatment status. However, Assumption 3 also allows for anticipation behavior, as long as we have a good understanding about the anticipation horizon δ . For instance, if units anticipate treatment by one period, Assumption 3 would hold with $\delta = 1$; see, e.g., [Laporte and Windmeijer \(2005\)](#) and [Malani and Reif \(2015\)](#) for the importance of accounting for potential anticipation behavior. Note that, under Assumption 3, $ATT(g, t) = 0$ for all pre-treatment periods such that $t < g - \delta$.

Next, we consider two alternative assumptions that impose restrictions on the evolution of untreated potential outcomes.

Assumption 4 (Conditional Parallel Trends based on a “Never-Treated” Group). *Let δ be as defined in Assumption 3. For each $g \in \mathcal{G}$ and $t \in \{2, \dots, \mathcal{T}\}$ such that $t \geq g - \delta$,*

$$\mathbb{E}[Y_t(0) - Y_{t-1}(0)|X, G_g = 1] = \mathbb{E}[Y_t(0) - Y_{t-1}(0)|X, C = 1] \text{ a.s..}$$

Assumption 5 (Conditional Parallel Trends based on “Not-Yet-Treated” Groups). *Let δ be as defined in Assumption 3. For each $g \in \mathcal{G}$ and each $(s, t) \in \{2, \dots, \mathcal{T}\} \times \{2, \dots, \mathcal{T}\}$ such that $t \geq g - \delta$ and $t + \delta \leq s < \bar{g}$,*

$$\mathbb{E}[Y_t(0) - Y_{t-1}(0)|X, G_g = 1] = \mathbb{E}[Y_t(0) - Y_{t-1}(0)|X, D_s = 0, G_g = 0] \text{ a.s..}$$

Assumptions 4 and 5 are two different conditional parallel trends assumptions that generalize the two-period parallel trends assumption to the case where there are multiple time periods and multiple treatment groups; see, e.g., [Heckman et al. \(1997, 1998\)](#), [Abadie \(2005\)](#) and [Sant’Anna and Zhao \(2020\)](#). Both assumptions hold after conditioning on covariates X . This can be important in many applications in economics particularly in cases where there are covariate specific trends in outcomes over time and when the distribution of covariates is different across groups. For example, [Heckman et al. \(1997\)](#) motivates conditional parallel trends assumptions in the context of evaluating a job training program. For evaluating job training programs, the distribution of observed covariates such as age, employment history, and years of education is often quite different between individuals who participate in job training and those that do not. When the path of labor market outcomes (in the absence of participating in job training) depends on these covariates, a conditional parallel trends becomes more plausible than an unconditional parallel trends assumption. In fact, ignoring the presence of covariate-specific trends can result in important biases when evaluating causal effects of policy interventions using unconditional DiD methods.

Assumptions 4 and 5 differ from each other depending on the comparison group one is willing to use in a given application. More specifically, Assumption 4 states that, conditional on covariates, the average outcomes for the group first treated in period g and for the “never-treated” group would have followed parallel paths in the absence of treatment. Assumption 5 imposes conditional parallel trends between

group g and groups that are “not-yet-treated” by time $t + \delta$.⁷ Importantly, both of these assumptions allow for covariate-specific trends and do not restrict the relationship between treatment timing and the potential outcomes, $Y_t(g)$ ’s. Thus, they are weaker than the randomization-based assumption made by [Athey and Imbens \(2018\)](#). We also note that the unconditional versions of Assumptions 4 and 5 are weaker than the parallel trends assumption imposed by [de Chaisemartin and D’Haultfoeuille \(2020\)](#) and [Sun and Abraham \(2020\)](#) as they impose fewer restrictions on the evolution of $Y_t(0)$ in pre-treatment periods; see, e.g., [Marcus and Sant’Anna \(2020\)](#) for a comparison.

In our view, practitioners may favor Assumption 4 with respect to Assumption 5 when there is a sizable group of units that do not participate in the treatment in any period, and, at the same time, these units are similar enough to the “eventually treated” units. When a “never-treated” group of units is not available or “too small”, researchers may favor Assumption 5 as it allows one to use more groups as valid comparison units, which potentially leads to more informative inference procedures. However, it is important to stress that favoring Assumption 5 with respect to Assumption 4 also involves potential drawbacks. For instance, in the absence of treatment anticipation ($\delta = 0$), Assumption 4 does not restrict observed pre-treatment trends across groups, whereas Assumption 5 does; see, e.g., [Marcus and Sant’Anna \(2020\)](#). Not restricting pre-treatment trends may be particularly meaningful in applications where the economic environment during the “early-periods” was potentially different from the “later-periods.” In these cases, the outcomes of different groups may evolve in a non-parallel manner during “early-periods”, perhaps because the groups were exposed to different shocks, while trends become parallel in the “later-periods.” We recommend taking these trade-offs into account when deciding which conditional parallel trends assumption is more appropriate for a given application.⁸

The final identifying assumption we impose is an overlap condition.

Assumption 6 (Overlap). *For each $t \in \{2, \dots, T\}$, $g \in \mathcal{G}$, there exist some $\varepsilon > 0$ such that $P(G_g = 1) > \varepsilon$ and $p_{g,t}(X) < 1 - \varepsilon$ a.s..*

Assumption 6 extends the overlap assumption in [Heckman et al. \(1997, 1998\)](#), [Abadie \(2005\)](#), and [Sant’Anna and Zhao \(2020\)](#) to the multiple groups and multiple periods setup. It states that a positive fraction of the population starts treatment in period g , and that, for all g and t , the generalized propensity score is uniformly bounded away from one. Assumption 6 rules out “irregular identification”, see, e.g., [Khan and Tamer \(2010\)](#).

Remark 1. *Note that Assumption 3 and Assumption 4 (Assumption 5) are intrinsically connected. For instance, when one imposes the “no-anticipation” condition (so that $\delta = 0$), Assumption 4 would then impose conditional parallel trends only for post-treatment periods $t \geq g$. If one allows for anticipation behavior (so that $\delta > 0$), Assumption 4 would then impose conditional parallel trends in some pre-treatment periods, too. In fact, the parallel trends assumptions become stronger as one increases δ . To*

⁷[Athey and Imbens \(2006\)](#) and [de Chaisemartin and D’Haultfoeuille \(2017\)](#) also consider using “not-yet-treated” units as comparison groups in related DiD procedures.

⁸It may be tempting to use statistical pre-tests to select between different versions of the parallel trends assumption. However, the results of [Roth \(2020\)](#) show that such a practice can lead to important distortions when conducting inference. Thus, we do not recommend following this path, but instead recommend taking the context of the application into account in order to choose the appropriate parallel trends assumption.

where $\widehat{C}(g, t)$ is as defined in Algorithm 1.

Remark 10. In DiD applications, it is common to use “cluster-robust” inference procedures; see, e.g., Wooldridge (2003) and Bertrand et al. (2004). However, we note that the choice of whether to cluster or not is usually not obvious, and depends on the kind of uncertainty one is trying to reflect; see, e.g., Abadie et al. (2017) for a discussion in a cross-sectional setup.¹⁵ In the case that one wishes to account for clustering to reflect “cluster-based” sampling uncertainty, we note that this can be done in a straightforward manner using a small modification of the multiplier bootstrap described above, provided that the number of cluster is “large.” More precisely, instead of drawing observation-specific V ’s, one simply needs to draw cluster-specific V ’s; see, e.g., Sherman and Le Cessie (2007), Kline and Santos (2012), Cheng et al. (2013), and MacKinnon and Webb (2018, 2020). If the number of clusters is “small,” however, the application of the aforementioned bootstrap procedure is not warranted.¹⁶

Remark 11. In Algorithm 1 we have required an estimator for the main diagonal of Σ . However, we note that if one takes $\widehat{\Sigma}(g, t) = 1$ for all (g, t) , the result in Corollary 1 continues to hold. However, the resulting “constant width” simultaneous confidence band may be of larger length; see, e.g., Montiel Olea and Plagborg-Møller (2018) and Freyberger and Rai (2018).

Remark 12. The above results focus on making inference about $ATT(g, t)$ ’s in (effective) post-treatment periods $t \geq g - \delta$. Although the limited anticipation condition in Assumption 3 implies that $ATT(g, t) = 0$ for all $t < g - \delta$ regardless of the group g , it is common practice to also estimate these pre-treatment parameters and use them to assess the credibility of the underlying identifying assumptions. Note that our DiD estimands (2.2) - (2.7) can be easily adjusted to include these by simply replacing the “long differences” $(Y_t - Y_{g-\delta-1})$ with the “short differences” $(Y_t - Y_{t-1})$ for all $t < g - \delta$. All our results continues to hold when one augments $\widehat{ATT}_{t \geq (g-\delta)}^{dr, nev}$ to also include these estimates for the $ATT(g, t)$ ’s in the pre-treatment periods $t < g - \delta$.

4.2 Asymptotic Theory for Summary Parameters

Assume, for simplicity, that Assumption 3 holds with $\delta = 0$. In this section, we discuss how one can estimate and make inference about the summary measures of the casual effects discussed in Section 3. More concisely, we consider parameters of the form of θ as defined in (3.1), which covers all of the aggregated parameters discussed in Section 3.

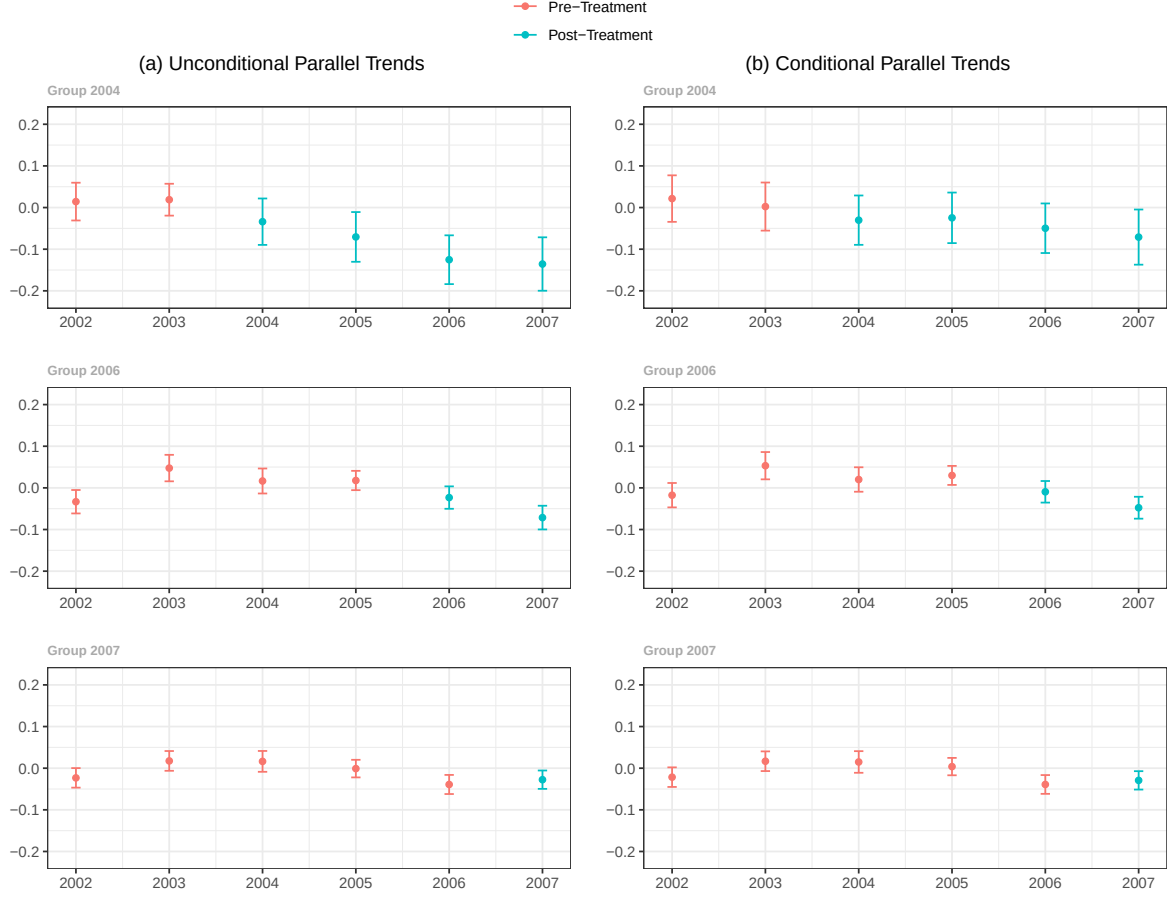
Given the discussion in Section 4.1, a natural way to estimate θ is to use the plug-in type estimators

$$\hat{\theta} = \sum_{g \in \mathcal{G}} \sum_{t=2}^{\mathcal{T}} \widehat{w}(g, t) \widehat{ATT}_{dr}^{nev}(g, t; 0),$$

¹⁵The formal results in Abadie et al. (2017) focus on the cross section case and rely on additional functional form restrictions that we do not impose in this paper. Fully extending the results of Abadie et al. (2017) to the semiparametric panel data case is beyond the scope of our paper.

¹⁶In such cases, provided that one is comfortable imposing additional functional form assumptions, one could use alternative procedures such as Conley and Taber (2011) and Ferman and Pinto (2019). Extending these proposals to our setup is beyond the scope of this paper though.

Figure 1: Minimum Wage Group-Time Average Treatment Effects



Notes: The effect of the minimum wage on teen employment estimated under the unconditional parallel trends assumption (Panel (a)) and the conditional parallel trends assumption (Panel (b)). Red lines give point estimates and uniform 95% confidence bands for pre-treatment periods allowing for clustering at the county level. Under the null hypothesis of the parallel trends assumption holding in all periods, these should be equal to 0. Blue lines provide point estimates and uniform 95% confidence bands for the treatment effect of increasing the minimum wage allowing for clustering at the county level. The top row includes states that increased their minimum wage in 2004, the middle row includes states that increased their minimum wage in 2006, and the bottom row includes states that increased their minimum wage in 2007. The estimates in Panel (b) use the doubly robust estimator discussed in the text.

increasing the minimum wage on teen employment appears to be negative and increasing in magnitude the longer states are exposed to the higher minimum wage. In particular, in the first year that a state increases its minimum wage, teen employment is estimated to decrease by 2.7%, in the second year it is estimated to decrease by 7.1%, in the third year by 12.5%, and in the fourth year by 13.6%. Notice that the last two dynamic treatment effect estimates are exactly the same as the estimates coming from Illinois alone because Illinois is the only state that is treated for at least two years. These results are robust to keeping the composition of groups constant by “balancing” the groups across different lengths of exposure to the treatment (see the row in Table 3 labeled ‘Event Study w/ Balanced Groups’). When we restrict the sample to only include groups that had a minimum wage increase for at least one full year (i.e., we keep groups 2004 and 2006 but not 2007), we estimate that the effect of increasing the minimum wage on impact is 2.7% lower teen employment and 7.1% lower teen employment one year after

on teen employment across all groups that increased their minimum wage is a 3.1% reduction in teen employment. This estimate is much different from the TWFE estimate. In addition, the pattern of dynamic treatment effects where the magnitude of the effect of increasing the minimum wage tends to increase with length of exposure is the same as in the unconditional case.

Overall, our results suggest that increasing the minimum wage decreased teen employment relative to what it would have been without the policy change. However, there are some important limitations of our application. First, some of the estimates of pseudo group-time average treatment effects in pre-treatment periods in Figure 1 are significantly different from zero which provides some suggestive evidence against the parallel trends assumption. Second, as discussed in the Supplementary Appendix, there is some heterogeneity in the size of the minimum wage increase itself across states which could complicate the interpretation of our results. Together, these suggest that our results should be interpreted with some caution. That being said, we think that the key takeaway from the application is that, (implicitly) holding the main identifying assumptions constant, in a prominent application in economics that has many very common features (treatment effect heterogeneity, dynamic effects, and staggered treatment adoption) the choice of estimation method can potentially lead to qualitatively different conclusions.

6 Conclusion

This paper has considered Difference-in-Differences methods in the case where there are more than two time periods and units can become treated at different points in time – a commonly encountered setup in empirical work in economics. In this setup, we have proposed group-time average treatment effects, $ATT(g, t)$, that are the average treatment effect in period t for the group of units first treated in period g . Unlike the more common approach of including a post-treatment dummy variable in a two-way fixed effects regression, $ATT(g, t)$ corresponds to a well defined treatment effect parameter. We also showed that once $ATT(g, t)$ has been obtained for different values of g and t , they can be aggregated into other parameters to more concisely summarize heterogeneity with respect to some particular dimension of interest (such as length of exposure to the treatment) or, alternatively, into a single overall treatment effect parameter. In addition, our approach is suitable (i) for cases where the parallel trends assumption holds only after conditioning on covariates, (ii) using different comparison groups such as the never-treated or not-yet-treated, and (iii) when units can anticipate participating in the treatment and may adjust their behavior before the treatment is implemented. We view such flexibility as an important component of our proposed methodology.

We also provided nonparametric identification results leading to outcome regression, inverse probability weighting, and doubly robust estimands. Given that our nonparametric identification results are constructive, we proposed to estimate $ATT(g, t)$ using its sample analogue. We established consistency and asymptotic normality of the proposed estimators, and proved the validity of a powerful, but easy to implement, multiplier bootstrap procedure to construct simultaneous confidence bands for $ATT(g, t)$. The computational costs of our approach are generally low, and code for implementing our approach is available in the R `did` package.

Finally, we applied our approach to study the effect of minimum wage increases on teen employment. We found some evidence that increasing the minimum wage led to reductions in teen employment. More